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*Normalizing Facial Recognition Technology and The End of Obscurity**

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ABSTRACT This article argues that facial recognition technology is the most dangerous surveillance tool ever invented. Given the unique threats this morally suspect tool poses to privacy, civil liberties, human flourishing, and democracy, the only appropriate response is a ban. To justify our position, we explain why facial recognition is distinctive among biometrics, clarify how even seemingly benign and positive uses of the technology can trigger dangerous normalization dynamics, and pinpoint why current United States laws (with reverberations in the EU's AI Act) are designed to accelerate a slippery slope that makes mass surveillance nearly inevitable. Our most fundamental contribution lies in demonstrating how the concept of "obscurity" connects all three arguments: facial recognition algorithms technologically eviscerate obscurity, normalization psychologically undermines it, and the law doctrinally abandons it - a perfect storm that only prohibition can stop.

KEYWORDS: Facial Recognition Technology, Obscurity, Privacy, Normalization

TABLE OF CONTENTS: 1. Introduction. – 2. Facial Recognition Technology: A Uniquely Powerful Surveillance Tool. – 3. Normalizing Facial Recognition Technology. – 4. How the Law Facilitates Normalizing Facial Recognition. – 5. What Should Be Done?. – 6. Conclusion.

1. *Introduction*

This article advances a radical argument: facial recognition technology should be banned. Not limited, not constrained by guardrails, but outright prohibited. No society is better off for having facial recognition technology in it.

We reach this dire conclusion by examining three interconnected issues. First, facial recognition is fundamentally different from every other surveillance tool. It exploits our faces - features that are difficult to hide or change and often define who we are. Combined with billions of photos already online, facial recognition technology enables anyone to identify and track strangers instantly, inexpensively, and on a massive scale.

Second, even seemingly good uses of facial recognition technology create a psychological trap. When we unlock phones with our faces or breeze through airport security, we grow comfortable with being scanned. Each use makes the next seem less invasive. We call this process "normalization," and contend it only moves in one direction - toward accepting more surveillance.

Third, U.S. privacy laws exacerbate this problem rather than alleviating it. They focus on a threshold of harm that ignores small but persistent invasions that add up over time. Most dangerously, these "privacy nicks" actually weaken legal protections as people get used to surveillance, creating a downward spiral where yesterday's outrage becomes today's reasonable take on normal daily life. Even the EU's AI Act, despite its ban on real-time biometric surveillance in public spaces, contains exceptions that risk normalizing the very practices they are designed to prohibit. These three forces - technology, psychology, and law - work together to destroy what we call "obscurity," a type of privacy that we take for granted, which is essential for civil liberties, human flourishing, and democracy.

We proceed by examining each force in

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detail. Section 2 explains what makes facial recognition uniquely dangerous. Section 3 reveals how normalization works. Section 4 shows how U.S. law fails. Section 5 considers legal remedies but shows only one response is truly justified: prohibition.

2. *Facial Recognition Technology: A Uniquely Powerful Surveillance Tool*

There is only one way to have an honest and realistic discussion about facial recognition technology. We must first admit that it is a uniquely powerful surveillance tool that poses grave privacy and civil liberties risks.[1] At its core, facial recognition technology is morally suspect because it is a perfect tool for oppression. Even this tool's seemingly benign and beneficial uses could normalize more invasive ones, tilting democracies, including the United States, toward authoritarianism.[2] What's more, Europe, which set a global example with its groundbreaking EU AI Act and general commitment to protecting human rights, could buckle under public pressure for algorithmically provided security and convenience.[3]

Most U.S. policymakers reject banning facial recognition outright.[4] Some endorse targeted prohibitions, like banning real-time remote biometric identification in publicly accessible spaces (e.g., one of the restrictions in the EU AI Act), but even the most robust prohibitions to date are porous with broad exemptions.[5] Similarly, forward-thinking scholars propose novel measures to limit AI-infused surveillance, including facial recognition technology, that could enhance crucial Constitutional protections, such as the Fourth Amendment. But there are always exceptions, especially where public safety is concerned.[6] And most consider sweeping proposals for an outright ban, including prohibiting the use of the tool at airports, concerts, stores, and even our phones, impractical and overzealous.

When most policymakers are presented with the rallying cry of "ban it!" they nearly cannot believe sensible people would take such an extreme and uncompromising position for a technology with so many seemingly desirable uses. Lawmakers (particularly in the U.S.) and industry representatives who reject outright prohibitions seem to commonly make two misguided assumptions: (1) all digital innovations should be encouraged; (2) the key to responsible innovation is to limit the harmful *uses* of tools.[7] This is the allure of "dual use technologies," those capable of harmful and beneficial uses.[8] As long as a tool

[1] E. Selinger and W. Hartzog, *The Inconsentability of Facial Surveillance*, in *Loyola Law Review*, vol. 66, 2019, 101.

[2] Adrienne de Ruiter introduced the concept of "morally suspect" to describe deepfake technology. De Ruiter claims that since there are some justified uses of deepfakes, the technology cannot be classified as intrinsically morally wrong. However, she notes that since deepfake technology "lends itself particularly well" to actions that violate fundamental moral norms," it should be classified in a manner that puts on high alert-which is to say, as "morally suspect." A. de Ruiter, *The Distinct Wrong of Deepfakes*, in *Philosophy & Technology*, vol. 34, 2021, 1311-1332, https://link.springer.com/article/10.1007/s13347-021-00459-2; W. Hartzog, E. Selinger and J. Gunawan, *Privacy Nicks: How the Law Normalizes Surveillance*, in *Washington University Law Review*, vol. 101, 2024, 717; E. Selinger and B. Leong, *The Ethics of Facial Recognition Technology*, in C. Véliz (ed.), *The Oxford Handbook of Digital Ethics*, Oxford, UK, Oxford University Press, 2023, https://academic.oup.com/edited-volume/37078/chapter-abstract/337809992?redirectedFrom=fulltext.

[3] Parliament and Council Regulation 2024/1689 laying down Harmonised Rules on Artificial Intelligence and amending Regulations (EC) No. 300/2008, (EU) No. 167/2013, (EU) No. 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 and EU Parliament-EU Council, Regulation of the European Parliament and of the Council laying down harmonized rules on artificial intelligence (14 May 2024).

[4] M. Fidler and G. Hurwitz, *An Overview of Facial Recognition Technology in the United States*, in R. Matulionyte and M. Zalnieriute (eds.), *The Cambridge Handbook of Facial Recognition in the Modern State*, Cambridge, UK, Cambridge University Press, 2024.

[5] Shubam, *EU AI Act Will Fail Commitment to Ban Biometric Mass Surveillance*, in *Reclaim Your Face* 2024, https://reclaimyourface.eu/eu-ai-act-will-fail-commitment-to-ban-biometric-mass-surveillance; Access Now, *The EU AI Act: A Failure for Human Rights, a Victory for Industry and Law Enforcement*, in *Access Now*, 2024, www.accessnow.org/press-release/ai-act-failure-for-human-rights-victory-for-industry-and-law-enforcement.

[6] A.G. Ferguson, *Facial Recognition and the Fourth Amendment*, in *Minnesota Law Review*, vol. 105, 2021, 1105 and 1132.

[7] See, *e.g.*, U.S. Senate Committee on Commerce, Science, and Transportation, *Winning the AI Race: Strengthening U.S. Capabilities in Computing and Innovation*, 8 May 2025, www.commerce.senate.gov/2025/5/winning-the-ai-race-strengthening-u-s-capabilities-in-computing-and-innovation_2 ("The way to beat China in the AI race is to outrace them in innovation, not saddle AI developers with European-style regulations").

[8] See, C. Kang, *OpenAI's Sam Altman Urges A.I. Regulation in Senate Hearing*, N.Y. Times, 16 May 2023, www.nytimes.com/2023/05/16/technology/opena

can do *some* good, why not keep it around?

Policymakers often try to regulate dual-use tools in a technology-neutral way. They believe passing tech-neutral rules like "do not lie" and "do not harm" that apply to all tools is the best way to discourage bad outcomes while encouraging good ones. To regulate facial recognition technology in a tech-neutral manner, lawmakers might assign risk tiers and impose stricter limitations on higher-risk activities. Under this approach, police officers would be able to use facial recognition technology to locate missing children, but stalkers would be barred from using facial recognition technology to harass others. Lawmakers could even pass some rules to ensure that tools like facial recognition technology work equally well for dominant groups and marginalized populations. Ultimately, the tech-neutral approach assumes that all innovation can be channeled for the public good and that, with proper oversight, the benefits of facial recognition technology, like every other digital tool, can outweigh the risks.

In many instances, it is a good idea to use tech-neutral regulation to get the most good from dual-use tools.[9] But facial recognition is no ordinary tool. It is the most dangerous surveillance tool ever created. No other surveillance tool rivals its affordances for destruction and oppression. Lawmakers seeking to preserve the benefits of facial recognition underestimate its core purpose and capabilities.

Given the power facial recognition bestows on the watcher, it stands a good chance of constantly drifting towards abuse, with its harms outweighing its benefits. Facial recognition technology deserves to be uniquely stigmatized because it is the most lethal combination of apparent convince bundled with hidden dangers imaginable. Our faces can reveal who we are, where we've been, and what we're doing more than any other single currently observable trait. Using computer vision and machine learning algorithms to automate the detective work of revealing who an unknown person is based on their most personal physiological features encourages heightened mass surveillance at an unprecedented scale and scope.[10]

As if this was not enough, facial recognition involves more than just identifying who people are. Because facial recognition is easy to implement inside existing surveillance systems, the threat facial recognition systems pose goes far beyond rapidly processing vast amounts of visual data from cameras and comparing detected faces against large databases of known individuals. Cameras with facial recognition capabilities can be easily and cheaply enhanced with related plug-and-play features. One additional tool is emotion recognition software that supposedly infers what people are feeling based on their facial expressions.[11] This type of information is known as facial characterization data, and it is used for junk science. Its misuse has been rightly compared to the discredited, eugenicist-approved, 19th-century practice of cranioscopy (later known as phrenology).[12] According to phrenology, you could learn about a person's character and mental abilities by studying the shape of different parts of their skull. Consider what could happen now with digital phrenology. If

i-altman-artificial-intelligence-regulation.html ("Tech companies have argued that Congress should be careful with any broad rules that lump different kinds of A.I. together. In Tuesday's hearing, Ms. Montgomery of IBM called for an A.I. law that is similar to Europe's proposed regulations, which outlines various levels of risk. She called for rules that focus on specific uses, not regulating the technology itself. "At its core, A.I. is just a tool, and tools can serve different purposes," she said, adding that Congress should take a "precision regulation approach to A.I.").

[9] See, *e.g.*, B.J. Koops, *Should ICT Regulation Be Technology-Neutral?*, in B.-J. Koops, M. Lips, C. Prins and M. Schellekens (eds.), *Starting Points for ICT Regulation. Deconstructing Prevalent Policy One-Liners*, vol. 9, The Hague, T.M.C. Asser Press, 2006, 77-108.

[10] J. Stanley, *The Dawn of Robot Surveillance*, in *American Civil Liberties Union*, 2019, www.aclu.org/publications/dawn-robot-surveillance. Institutions have also begun to take notice. See Artificial intelligence must be grounded in human rights, says High Commissioner, United Nations (12 July 2023), www.ohchr.org/en/statements-and-speeches/2023/07/artificial-intelligence-must-be-grounded-human-rights-says-high.

[11] N. Andalibi, *Emotion-tracking AI on the job: Workers fear being watched – and misunderstood*, in *The Conversation*, 6 March 2024, https://theconversation.com/emotion-tracking-ai-on-the-job-workers-fear-being-watched-and-misunderstood-222592, ("Emotion artificial intelligence uses biological signals such as vocal tone, facial expressions and data from wearable devices as well as text and how people use their computers, promising to detect and predict how someone is feeling. It is used in contexts both mundane, like entertainment, and high stakes, like the workplace, hiring and health care").

[12] L. Stark and J. Hutson, *Physiognomic Artificial Intelligence*, in *Fordham Intellectual Property, Media & Entertainment Law Journal*, vol. 32, 2022, 922.

police used body cameras with facial and emotional recognition, non-threatening suspects could easily be labeled aggressive. Given pervasive racial, gender, and ability biases, it would not be surprising if those who get misclassified are members of marginalized communities and are at risk of being subject to overly forceful responses.

To better explain why we think facial recognition technology is uniquely dangerous, let us compare it to other combinations of artificial intelligence (AI) and surveillance, like automated license plate readers or social media monitoring systems. These technologies also have profound implications for privacy and power. However, not everyone is on social media or drives a car - and even those who are and do spend time away from them. But we bring our unique faces with us everywhere we go. Our faces are constant, personally identifiable features that we cannot easily change, often cannot conceal without arousing suspicion, and link our online and offline lives. They are also core to our identity. When you think of a friend in your mind, you don't think of their elbow or shoulder. You certainly don't think of their fingerprints or palm prints. You think of their face. Unlike other biometric identifiers, our faces are also already scattered all over the Internet, making turnkey surveillance easy and cheap. No other surveillance system - not license plate readers, CCTV cameras, social media, or any other tool - comes close to that blend of vulnerabilities in terms of depth, breadth, and cost efficiency.

Consider I-XRAY, a recent activist project from Harvard students AnhPhu Nguyen and Caine Ardayfio.[13] They demonstrated how readily available technology and services - Ray-Ban Meta smart glasses (to photograph a face), a public facial recognition search engine (Pimeyes), easily obtainable information from data brokers, and a large language model (that can rapidly pick out and extract personal information from unstructured data) - can produce horrifying results. As the students walked around in public secretly photographing strangers, they nearly instantaneously identified their victims. While doing so, an app provided them with a wealth of information about the strangers and, in some cases, their family members. This example of hyper-efficient, unauthorized, and hidden doxxing shows just how thin the line is between living in a society with adequate obscurity protections and creating one where your life is an open book.[14]

"Obscurity," as we use the term here, is a concept similar to "privacy." However, obscurity has distinctive features that make it a better term for understanding why facial recognition technology is so menacing.[15] Obscurity refers to the practical challenges, also known as transaction costs, associated with finding or understanding information. Unlike secret or truly anonymous information, obscure information may *literally* be available to a member of the public with sufficient motivation, resources, and know-how. But obscure information is, as a practical matter, difficult enough to find or use that it's likely to remain private. While not a perfect safeguard, obscurity thrives on the friction created by time, effort, and other deterrents. These expenses provide a buffer against causal and arbitrary intrusions into our lives.

Obscurity allows us to have something like group privacy and privacy in public.

Imagine eating at a restaurant and two strangers, who are not public figures, are having an intimate, quiet conversation about their personal lives at another table. They will be protected by a natural zone of obscurity that prevents most people from trying to hear much of what they have to say. Of course, determined intruders will pose a threat. A nosy person might be a problem, and a gossip who has the rare skill of being able to read lips

[13] J. Cox, *Someone Put Facial Recognition Tech onto Meta's Smart Glasses to Instantly Dox Strangers*, in *404 Media*, 2024, www.404media.co/someone-put-facial-recognition-tech-onto-metas-smart-glasses-to-instantly-dox-strangers; K. Hill, *Two Students Created Face Recognition Glasses. It Wasn't Hard.*, in *New York Times*, 2024, www.nytimes.com/2024/10/24/technology/facial-recognition-glasses-privacy-harvard.html.

[14] This is to say nothing of the risk of fraud and identity theft, which becomes irreversible when biometric data is stolen or exposed in a data breach.

[15] See, *e.g.,* W. Hartzog and E. Selinger, *Surveillance as Loss of Obscurity*, in *Washington and Lee Law Review*, vol. 72, 2015, 1343-46 [hereinafter Hartzog and Selinger, Surveillance]; W. Hartzog and E. Selinger, *Increasing the Transaction Costs of Harassment*, in *Boston University Law Review Annex*, vol. 95, 2015, 47; E. Selinger and W. Hartzog, *Obscurity and Privacy*, in , J. Pitt and A. Shew (eds.), *Spaces for the Future: Routledge Companion to Philosophy of Technology,* London, Routledge, 2018, www.routledge.com/Spaces-for-the-Future-A-Companion-to-Philosophy-of-Technology/Pitt-Shew/p/book/9780415842969; see also W. Hartzog and F. Stutzman, *The Case for Online Obscurity*, in *California Law Review*, vol. 101, 2013, 1, especially 5.

is especially dangerous. However, most people will go about their lives without bothering to find out who is at the table and why they are having a particular discussion. They lack the incentive to divert their attention, and the effort required to snoop is better spent elsewhere.

Contrast this scenario with the same two strangers wanting to have the same conversation while strolling through a smart city park where surveillance cameras are plastered everywhere. Furthermore, imagine the cameras linked to the same technologies that were used in the I-XRAY demonstration, along with emotion recognition and lip-reading software, which are easy add-ons. Given the EU AI Act, the scenario is more likely to occur in the U.S. than in Europe. However, as our discussion of normalization will suggest, things may change.

The mere presence of this infrastructure could easily have chilling effects that prevent people from speaking their minds.[16] After all, there will be a dramatic reduction in the transaction costs of surveillance and data processing, making it easier and more likely for public spaces to be constantly watched and analyzed. The reasonable expectation of obscurity - and the freedom it provides -would be eroded by the constant potential for observation and analysis. Chilling effects would occur whether or not anyone actively monitors the data in real-time. The mere possibility is destabilizing.

These examples highlight two important things. First, it is easy to take obscurity for granted. For much of history, obscurity protections have not required legal measures; technological limitations have sufficed. Consider this analogy. Today, most people are not worried about the government reading their minds. Indeed, it would seem paranoid to believe the FBI or CIA has access to secret, perfected telepathy machines. And yet, not too long out, it would have seemed far-fetched to imagine government agents would have access to technology that could identify us by our faces wherever we go. And neurotechnology is getting better by the day.

Consequently, prominent neurotechnology scholars believe we should enact laws as quickly as possible to better protect our right to mental freedom, also known as cognitive liberty.[17] They recognize that people have mistaken the contingent biological safeguards protecting our private thoughts for absolute ones. The better we appreciate obscurity, the better prepared we will be to face future surveillance challenges. Conversely, if we had cared more about obscurity in the past, facial recognition technology might not be widely used today.

Second, facial recognition differs considerably from much more easily regulated biometrics, like fingerprinting. Fingerprinting typically requires some form of physical interaction, like touching a scanner or leaving a visible print on objects, even if only a latent one. In contrast, facial recognition can be performed at a distance on vast numbers of people without anyone giving even nominal consent or being aware of the collection. Of course, contactless fingerprint scanning exists. However, it is not the typical approach to acquiring fingerprints, and the process is limited to short distances away. Even as fingerprinting research advances, the fact remains that facial recognition poses a much more immediate threat, which is why it requires the greatest restraints.

3. *Normalizing Facial Recognition Technology*

One key to understanding how seemingly safe and positive uses of facial recognition are dangerous lies in how society comes to normalize practices. Generally, to say that something is "normal" is to describe it as either commonplace or adhering to an ideal or expected standard. Thus, the process of normalization involves repeating instances of something so frequently that people come to view it as commonplace or become acclimated to the occurrence such that it is no longer noteworthy.

Scholars across disciplines study the concept of normalization (including psychologists, sociologists, cognitive scientists, philosophers, and surveillance scholars) and privacy activists constantly highlight its dangers.[18] Normalizing facial

[16] See, J.W. Penney, *Understanding Chilling Effects*, in *Minnesota Law Review,* vol. 106, 1451, 2022; M. Buchi, N. Festic and M. Latzer, *The Chilling Effects of Digital Dataveillance: A Theoretical Model and an Empirical Research Agenda*, in *Big Data society*, Vol. 9, Issue 1, 2022, 14, spec. 4.

[17] N. Farahany, *The Battle for Your Brain: Defending the Right to Think Freely in the Age of Neurotechnology*, 2023.

[18] See Z. Bauman and D. Lyon, *Liquid Surveillance: A Conversation*, Cambridge, UK, Polity Press, 2012; J.B.

recognition systems is dangerous for several reasons. First, when people become acclimated to the privacy encroachments enabled by facial recognition, they become subject to a constant infliction of autonomy harms that fail to meet the harm thresholds demanded by privacy rules or prompt a social outcry. Second, normalization dynamics lead society to constantly re-negotiate its collective sense of reasonable expectations of privacy regarding when they can be identified and how much of their lives are an open book.[19] Our threshold for rejecting invasive new facial surveillance practices is perpetually being redrawn, excusing evermore invasive practices.[20] Without intervention, the normalization of facial recognition systems will result in a slow, inevitable, and irreversible erosion of obscurity and loss of power through exposure.

It's easy to imagine society completely normalizing facial recognition tools. Just look at what we've already gotten used to. We've long stopped noticing many surveillance tools like CCTV as exceptional or out of place.[21] Facial recognition is being sold to us as something convenient and desirable, and we're hastening our new normal of exposure with every new sensor we deploy in public, bring into our home, strap on our face, and put in our pocket.[22] Chris Gilliard calls this "luxury surveillance," and it will be our undoing.[23]

When facial recognition goes unchecked, profound societal ramifications can follow. People often develop positive beliefs about these tools that lead them to lose sight of how and why privacy protections provide essential checks against power. Normalizing facial recognition contributes to a "slippery slope dynamic," where society slides further and further into a state of diminishing privacy expectations. Empirical "slippery slope" arguments, the idea that a course of action will eventually snowball into unacceptable outcomes, are often presented as fallacious. Indeed, sometimes they are. The standard objection to empirical slippery slope prognostics is that they fail to identify credible causal mechanisms powerful enough to lead society towards ruinous outcomes without people and institutions changing course in time to avoid catastrophe. The skeptical objection thus suggests if society is ever heading in the wrong direction because of slippery slope factors like path-dependency, emergent governance responses will kick in and prevent tragedy. However, not all slippery slope arguments are fallacious, which is why we argue that slippery slope dynamics have gotten a bad rap. They are merely claims about expected outcomes, which can be prevented if the power of the slippery slope mechanisms can be muted. Thus, to make a valid empirical slippery slope argument, one must specify the causal mechanisms and

Rule, *Private Lives and Public Surveillance: Social Control in the Computer Age*, Belmont, CA, Wadsworth Publishing, 1974, 22; S.E. Igo, *The Known Citizen: A History of Privacy in Modern America*, Cambridge, MA, Harvard University Press, 2018; O.H. Gandy, Jr., *The Panoptic Sort: A Political Economy of Personal Information*, 2nd ed., Cambridge, MA, MIT Press, 2019, 31; D. Lyon, *Surveillance Studies: An Overview*, Cambridge, UK, Polity Press, 2007, 27; G.T. Marx, *Windows into the Soul: Surveillance and Society in an Age of High Technology*, Chicago, University of Chicago Press, 2016; W.G. Staples, *Everyday Surveillance: Vigilance and Visibility in Postmodern Life*, 2nd ed., Boulder, CO, Paradigm Publishers, 2013, 5; see also: S. Byrne, *The Banality of Surveillance*, in *Surveillance & Society*, vol. 20, 2022, 372; D.M. Wood and K. Ball, *Brandscapes of Control? Surveillance, Marketing and the Co-Construction of Subjectivity and Space in Neoliberal Capitalism*, in *Marketing Theory*, vol. 13, 2013, 47; G. Deleuze, *Postscript on the Societies of Control*, in *October*, vol. 59, 1992, 3.

[19] This is particularly consequential in the United States which conditions many of its privacy rules upon a finding of a "reasonable expectation of privacy." See, W. Hartzog, E. Selinger and J. Gunawan, *Privacy Nicks: How the Law Normalizes Surveillance*, in *Washington University Law Review*, vol. 101, 2024; E. Selinger and J.H. Rhee, *Normalizing Surveillance*, in *North European Journal of Philosophy*, vol. 22, 2021, 49.

[20] W. Hartzog, E. Selinger and J. Gunawan, *Privacy Nicks: How the Law Normalizes Surveillance*, in *Washington University Law Review*, vol. 101, 2024; E. Selinger and J.H. Rhee, *Normalizing Surveillance*, in *North European Journal of Philosophy*, vol. 22, 2021, 49.

[21] R. Santana and R. Gentilo, *TSA Is Testing Facial Recognition at More Airports, Raising Privacy Concerns*, in *AP News*, 2023, https://apnews.com/article/facial-recognition-airport-screening-tsa-d8b6397c02afe16602c8d34409d1451f; B. Goold *et al., The Curious Case of Surveillance Cameras*, in *British Journal of Criminology*, vol. 53, 2013, 977.

[22] See, *e.g.*, S. Andrade, *Clear Wants to Scan Your Face at Airports. Privacy Experts Are Worried.*, in *Washington Post*, 2023, www.washingtonpost.com/travel/2023/12/20/clear-facial-recognition-technology-airport-security/; J. Winner, *3 Tangible Ways That AI Will Continue to Make Your Life Better*, in *Fast Company*, 2023, www.fastcompany.com/90892907/3-tangible-ways-that-ai-will-continue-to-make-your-life-better.

[23] C. Gilliard, *Amazon and the Rise of 'Luxury Surveillance'*, in *The Atlantic*, 2022, www.theatlantic.com/technology/archive/2022/10/amazon-tracking-devices-surveillance-state/671772.

explain why their influence is not likely to be dampened adequately in time to prevent disaster.

Scholars have identified two processes by which initially "creepy" deployments of technology become normalized.[24] We think facial recognition systems will follow suit. A process called "unexceptional habituation" occurs when "people in liberal Western democracies take ubiquitously encountered surveillance systems for granted - seeing them as so commonplace and mundane they are not worth thinking about critically."[25] The more a tool is deployed, the less remarkable it becomes as it fades into the background. For example, most people probably use Apple's FaceID to unlock their phones without much critical reflection. They also scan their faces to board airplanes and get into concerts. While most people's first encounter with these facial recognition technologies might have felt novel, each encounter is less remarkable than the last because of people's growing familiarity with the tool. Ultimately, people will stop critically engaging with these systems because they are part of their habits. Even if uses like FaceID or airport identity verification are themselves relatively innocuous, they habituate people to technologies that are ready-made for more obviously harmful surveillance programs.

Once people come to see facial recognition as exceptional, they will often come to view a practice as acceptable, if not desirable, reflecting a psychological dynamic called "favorably disposed normalization."[26] The idea is that people often take moral cues from the behavior of others, so observing routine behavior involving facial recognition systems could signal that the technology is good.[27] There is also evidence that people come to rationalize their own use of a technology as desirable to avoid the difficult conclusion they are acting wrongfully.[28] Few like to think of themselves as the bad person, so it's often just easier to recast our use of dangerous tools as benign so we can continue to see ourselves as good.[29] Or maybe what happens when we come to view a practice as acceptable is that we look back on our past behavior, such as using Face ID to open our phones, to retroactively attribute those actions to an attitude or motive, such as thinking facial recognition is efficient and helpful.[30]

Another driver normalizing facial recognition is people's tendency to selectively focus on its benefits and overlook or unduly discount its harms. Consider the example of airport screening, which offers an allegedly fast and convenient way to confirm someone's identity. In the moment when a person stands in line for screening, the benefit of arriving at the airport gate on time is clear and easy to prioritize. But the deeper harms of facial surveillance - such as having their faceprint potentially stored indefinitely in a government database - are abstract, if not completely invisible.[31] Because of this, even when people are aware that they have the option to avoid facial recognition technology, most do not opt out.[32]

[24] E. Selinger and J.H. Rhee, *Normalizing Surveillance*, in *North European Journal of Philosophy*, vol. 22, 2021, 49.
[25] *Id*.
[26] *Id.*
[27] *Id.*
[28] See, *e.g.*, J.P. Friesen *et al*., *System Justification: Experimental Evidence, Its Contextual Nature, and Implications for Social Change*, in *British Journal of Social Psychology*, vol. 58, 2019, 315.

[29] People generally are motivated to see themselves positively, as moral, intelligent, and in control of their lives. To maintain this narrative and minimize inconsistency when making decisions that seem unethical, stupid, or unfree, they often subconsciously turn to rationalization. Put otherwise, being aware of a gap between how we would like to act and how we actually behave can be stressful because it creates cognitive dissonance. Rationalization is ameliorative because it can minimize or dispel cognitive dissonance. Rationalization provides people with a means to convince themselves they should see their situation differently - that seemingly troubling behavior is justifiable, tolerable, and in some cases, even laudable. J.P. Friesen *et al*., *System Justification: Experimental Evidence, Its Contextual Nature, and Implications for Social Change*, in *British Journal of Social Psychology*, vol. 58, 2019, 315; L. Festinger, *A Theory of Social Comparison Processes*, in *Human Relations*, vol. 7, 1957, 117.
[30] D.J. Bem, *Self-Perception Theory*, in *Advances in Experimental Social Psychology*, vol. 6, 1972, 1.
[31] See, *e.g.*, C. Liu, *The Federal Government Just Can't Get Enough of Your Face*, in *Electronic Frontier Foundation*, 15 September 2021, www.eff.org/deeplinks/2021/09/federal-government-just-cant-get-enough-your-face; J.Reid, TSA Starts Testing Facial Recognition Technology at 16 Major Airports, Avionics International (29 December 2022), www.aviationtoday.com/2022/12/29/tsa-starts-testing-facial-recognition-technology-16-major-airports.
[32] See, A. Funk, *I Opted Out of Facial Recognition at the Airport—It Wasn't Easy*, in *Wired*, 2 July 2019, www.wired.com/story/opt-out-of-facial-recognition-at-the-airport/ ("Federal agencies and airlines claim that facial recognition is an opt-out system, but my recent experience suggests they are incentivizing travelers to have their faces scanned - and disincentivizing them to sidestep the tech - by not clearly communicating

As a result, giving people a "choice" regarding facial recognition is another way to normalize it. On the surface, allowing people to decide for themselves whether to use facial recognition seems empowering and autonomy-enhancing. In reality, however, those who are reluctant to undergo face scans don't have a real choice to opt out of facial recognition technology. Opting out would mean they would have to go through a separate screening procedure that is likely slower and inconvenient.[33] People would view this as a penalty, especially because the vast majority of travelers prefer to go through security as fast as possible.[34] The normalizing effects of strategies designed to give people control over personal information are just another reason to be skeptical of "consent and control" schemes to protect people's privacy.[35]

To be sure, the legitimacy and presence of these normalizing effects are still open for debate.[36] But they are so intuitively plausible that academics and activists frequently offer normalization warnings like the following ones: if surveillance intensifies at schools, students will be more inclined to accept more intrusive instances of it later in life; and, if during emergencies, new forms of surveillance get introduced, citizens will be more willing to look favorably upon comparable, if not more expansive varieties, after the crises end.[37] In other words, the history of surveillance shows that it only runs one way: more.[38] Facial surveillance will be no different. We will get used to it and, as a result, view it as desirable.[39]

Normalizing facial recognition will cost society dearly. We will end up entrenching these systems slowly and incrementally, ensuring that society does not focus enough critical attention to mount meaningful democratically driven resistance. As such, our expectations of privacy will be perpetually eroded until we reach a fully transparent society. People living in that future will be deprived of crucial avenues for human flourishing. Consequently, failing to take facial recognition seriously is fundamentally a problem of intergenerational justice that ensures future generations will have little to no obscurity.

4. *How the Law Facilitates Normalizing Facial Recognition*

Lawmakers should act quickly and decisively to address the threat that normalizing facial recognition poses to our obscurity. Though industry, lawmakers, advocates, and scholars disagree about the substance and the goals of facial recognition regulation, the common wisdom is that robust new privacy rules should preserve or at least re-establish our solitude and freedom in light of invasive facial recognition practices.[40]

alternative options. Last year, a Delta customer service representative reported that only 2 percent of customers opt out of facial recognition. It's easy to see why.").

[33] A. Funk, *I Opted Out of Facial Recognition at the Airport—It Wasn't Easy,* in *Wired*, 2 July 2019, www.wired.com/story/opt-out-of-facial-recognition-at-the-airport.

[34] This is an instance of 'domination' - a form of exercising power through control. In this particular situation, control is exercised by giving people the illusion of freedom, which encourages the continued use of facial recognition technology. As demonstrated here, normalization is in many ways enabled by the processes of domination.

[35] E. Selinger and W. Hartzog, *The Inconsentability of Facial Surveillance*, in *Loyola Law Review*, vol. 66, 2019, 101; D. Solove and W. Hartzog, *Kafka in the Age of AI and the Futility of Privacy as Control*, in *Boston University Law Review*, vol. 104, 2024, especially 1025-1029.

[36] Additional research is needed to better understand the psychology behind normalization in a privacy context. What is also clear, however, is that the law cannot wait for more research before regulating facial recognition technology. In our view, the various ways surveillance can be normalized are enough to show that the absence of regulation will create the conditions for a slippery slope. When left unchecked, the processes of normalization may undermine democracy and our fundamental rights, including autonomy.

[37] E. Selinger and J.H. Rhee, *Normalizing Surveillance*, 49.

[38] See D. Lyon, *Surveillance: A Short Introduction,* 2024.

[39] Research suggests that the more common something is, the more likely people are to see it as desirable. Simply "increasing the frequency of something occurring," such as surveillance more becoming more prevalent, can lead people to perceive it as "more normal," not just increasingly widespread. Given the practical value of heightened moral motivation for rectifying injustice, in some circumstances, "beliefs about normality might be more important than moral beliefs." E. Selinger and J.H. Rhee, *Normalizing Surveillance* at 62. Supporting evidence for this thesis exists in the experimental literature on environmental messaging. N.J. Goldstein *et al.*, *A Room with a Viewpoint: Using Social Norms to Motivate Environmental Conservation in Hotels*, in *Journal of Consumer Research*, vol. 35, 2008, 472; R.B. Cialdini *et al.*, *Managing Social Norms for Persuasive Impact*, in *Social Influence*, vol. 1, 2006, 3.

[40] See H. Rahnama and A. Pentland, *The New Rules of Data Privacy*, in *Harvard Business Review*, 2022, https://hbr.org/2022/02/the-new-rules-of-data-privacy. See generally O.S. Kerr, *The Fourth Amendment and*

Scholars sometimes frame proposals for new facial recognition rules and other checks on surveillance in terms of re-establishing or preserving our state of privacy.[41] The idea is to keep change steady without too many wild swings toward exposure or opacity. Unfortunately, under our current surveillance frameworks, equilibrium adjustment for facial recognition is impossible. Even our most robust privacy laws will normalize facial recognition in ways that will leave society worse off due to the erosion of obscurity.[42] Lawmakers concerned about the risks of facial recognition have so far largely failed to see the big picture.[43]

Obscurity, perhaps more than any other conceptualization of privacy, is jeopardized by the normalization of facial recognition technology. It feels mundane because it exists all around us. We are, by default, obscure to the world unless something happens to make us more apparent, like facial recognition tools finding and highlighting an obscure photo of us on the Internet or identifying us while out and about in public. So, it's unsurprising that obscurity is already one of the law's least protected notions of privacy. Society largely takes it for granted. But something worse is actually happening under the radar. Our laws don't just ignore obscurity diminutions that happen incrementally and over time. They *facilitate* the normalization of tools and practices that erode obscurity. Such facilitation occurs through four legal dynamics that were established in technologically simpler times long before facial recognition technology existed. These dynamics amplify the normalization processes we have discussed, making social and psychological processes like favorably disposed normalization and unexceptional habituation more likely to take hold.

First, U.S. privacy law rigidly demands demonstrable and significant privacy harms to justify legal intervention. Privacy law suffers from a limited vocabulary to differentiate harms based on their magnitude. Under the law, people usually either suffer a privacy violation or they don't. But that's not how people experience privacy incursions, particularly obscurity erosions.[44] Some harms, like those that result in extreme emotional distress, debilitating physical injury, and deprivation of significant life opportunities, clearly are worse than mild annoyances and feelings of "creepiness." But over time, even dispersed and incremental harms take their toll. By ignoring small, *de minimus* encroachments (what we call "privacy nicks"), U.S. lawmakers encourage the normalization of harmful extractive and exploitative practices using surveillance tools like facial recognition.

Privacy nicks are the proverbial "thousand cuts" that lead to death, which explains why even robust privacy protections have failed to halt the expansion of facial surveillance.[45] Privacy nicks are caused by the deployment of new information technologies like facial recognition that generally seem tolerable but can lead to perilous long-term individual and social consequences. They often fail to raise social alarms or trigger legal privacy protections. And they are enabled by the proliferation of facial recognition affordances as part of our cameras and biometric sensors on doorbells, glasses, and watches, as well as the drift of surveillance and data analytics into new areas of our lives like travel, exercise, and social gatherings.

Next, the law over-endows the concept of waiver, particularly in U.S. frameworks. The law typically justifies otherwise objectionable behavior like facial surveillance when people consent to data practices or voluntarily expose themselves to others. This is a dangerous approach for facial recognition because our faces are the most exposed part of our bodies and the most salient physical tie to our

New Technologies: Constitutional Myths and the Case for Caution, in *Michigan Law Review*, vol. 102, 2004, 801, especially 855-57.

[41] O.S. Kerr, *An Equilibrium-Adjustment Theory of the Fourth Amendment*, in *Harvard Law Review*, vol. 125, 2011, 476, especially 480.

[42] W. Hartzog, E. Selinger and J. Gunawan, *Privacy Nicks: How the Law Normalizes Surveillance*, in *Washington University Law Review*, vol. 101, 2024; E. Selinger and J.H. Rhee, *Normalizing Surveillance*, in *North European Journal of Philosophy*, vol. 22, 2021, 49.

[43] Though there are notable exceptions. See, CNIL, Facial Recognition: For a Debate Living Up to the Challenges, www.cnil.fr/sites/cnil/files/atoms/files/facial-recognition.pdf ("Experimentations should not have the ethical purpose or effect of accustoming people to intrusive surveillance techniques, with the more or less explicit aim of preparing the ground for further deployment. At this stage, there should be no question of making devices, which weaken people's autonomy or violate their fundamental rights, "acceptable").

[44] W. Hartzog and E. Selinger, *Surveillance*.

[45] See W. Hartzog, E. Selinger and J. Gunawan, *Privacy Nicks*, 717.

identities. One of the central assumptions behind the notion that people lack a reasonable expectation of privacy when in public is that they have consciously waived privacy protections by choosing to make their presence and activities visible to others. Judges considering privacy tort claims have said for years that there is no privacy in public information or people's whereabouts in public. Their opinions are littered with statements saying people have no legitimate expectation of privacy in the materials intended for publicly accessible websites. When people walk outside with their face exposed or select a photo of their face as their profile photo on a social media account, they likely are counting on a reasonable amount of obscurity in their whereabouts and a relatively transient or limited audience online.[46] Yet, in the U.S., lawmakers and judges routinely justify the scraping of profile photos to power facial surveillance systems, and when the law applies such a waiver rationale to justify surveillance and data processing, it presumes such actions are no longer worthy of additional scrutiny or restrictions within the current context.[47] This conclusion allows facial surveillance to run rampant and become normalized.

Third, privacy law has a misplaced focus on proximity, looking only at localized harms that imminently flow from the actions of others. When lawmakers and judges ask what the harm is from facial recognition, they often look to discrete acts of face scanning and identification to determine if a person suffered physical, financial, emotional, reputational, discrimination, or autonomy harm. This isolated focus on atomistic harms excludes scrutiny of the cumulative effects of discrete actions, thereby abdicating responsibility for addressing the costs of privacy-diminishing externalities. It's hard to hold many different actors responsible for the net result of a slow accretion of diffuse exposures. Who's to say that one scan of your face by an IoT doorbell is the one that resulted in a chilling effect or loss of autonomy? Privacy's obsession with proximity also includes another pathology: most privacy laws are self-oriented, almost to the point of narcissism. Any aspect of privacy law focused on "individual control over data" or "informational self-determination" is designed to force people to contemplate questions like "what is in it for me?" and "what is the worst that can happen to me or my data?" This egoistic bias ignores how one person's choices affect others. The result is the systemic oppression of marginalized people. In our current system, people of color, members of the LGBT+ community, and other marginalized people fall outside the scope of the majority's self-interested privacy considerations.

Finally, the U.S. rules that ostensibly keep facial recognition systems (and other surveillance practices) in check "too often look to people's expectations to set the limits of surveillance; yet over time, people become increasingly acclimated to being watched. People's desensitization to exposure affects how they view reasonable surveillance measures and fair tradeoffs."[48] Because legal thresholds in the U.S. are keyed to people's expectations, the normalization of privacy nicks results in a constant re-negotiation of privacy standards to our disadvantage. Without a firm backstop, nothing can prevent the gradual tolerance of a maximally transparent culture. It is already happening - slowly but surely.[49] As a result, we are lowering our 'reasonable expectations of privacy.' In sum, privacy law permits whatever people can be conditioned to tolerate. We are on track to tolerate everything.

5. *What Should Be Done?*

What should be done? First, we must recognize that the mere existence of facial recognition systems, which are often invisible, harms civil liberties because people will act differently if they suspect they're being surveilled. Even legislation that holds out the promise of stringent protective procedures won't prevent chill from impeding crucial

[46] E. Selinger and W. Hartzog, *Obscurity and Privacy*, in J. Pitt and A. Shew (eds.), *Spaces for the Future: Routledge Companion to Philosophy of Technology*, London, Routledge, 2018, www.routledge.com/Spaces-for-the-Future-A-Companion-to-Philosophy-of-Technology/Pitt-Shew/p/book/9780415842969; see also W. Hartzog and F. Stutzman, *The Case for Online Obscurity*, in *California Law Review*, vol. 101, 2013, 1, especially 5.

[47] D. Solove and W.Hartzog, *The Great Scrape: The Clash Between Scraping and Privacy,* in *California Law Review*, forthcoming, at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4884485.

[48] W. Hartzog, E. Selinger and J. Gunawan, *Privacy Nicks*, 717.

[49] *Id.*

opportunities for human flourishing by dampening expressive and religious conduct. Even requiring a warrant before facial recognition can be used as part of a government search will set the conditions for surveillance to occur, which will normalize tracking and identification, reorganize and entrench organizational structure and practices, and drive government and industry investment in facial recognition tools and infrastructure.

But is there a way to avoid normalizing facial recognition technology and protect obscurity? We think only outright prohibitions on the most dangerous surveillance tool ever will suffice. Nevertheless, recognizing that the most robust prohibitions might require time and political capital, we propose a few half-measures that can be applied to facial recognition but also be adapted more broadly to correct the law's normalization of surveillance. First, instead of focusing on individuals, our facial recognition rules should focus more on the collective good.[50] Privacy law is largely built around protecting individual autonomy and individual rights that individuals can exercise one right at a time. As we explained above, this "proximity" frame fails to consider the impact of surveillance on groups or society. Privacy law misses the forest by focusing only on the trees. This myopia causes lawmakers to miss some of the most harmful aspects of surveillance, including how coercion and discrimination only become apparent at scale. Focusing on the individual effect of surveillance also ignores how one person's actions can affect other people.[51]

As a first step away from individuals toward groups, lawmakers could better recognize threats to collective and social well-being as a privacy harm to satisfy damage and standing requirements.[52] They could also abandon the "reasonable expectation of privacy" test to focus on collective well-being or unjust uses of power, similar to calls for data collectors to be bound by duties of loyalty to trusting parties.[53] Dislodging individual expectations and individual harm as the center of privacy law would guide lawmakers to systematically examine the danger of privacy nicks.

Next, lawmakers should target the design of facial recognition systems themselves. Most privacy rules target surveillance and data processing behavior but are agnostic about the tools used to observe and collect our personal information. For example, electronic surveillance law prohibits the interception of aural signals or information but ignores how spycams hidden in everyday objects practically encourage surreptitious monitoring. Privacy torts limit the ways in which people can disclose private facts or intrude upon our seclusion, but they ignore how facial recognition tools make these actions so easy. It's a mistake for lawmakers to ignore the design of facial recognition technologies. Design is everywhere, design is power, and design is political.[54] When lawmakers ignore the design of information technologies, as they traditionally have in the U.S., they allow companies to escape accountability for malicious and negligent design decisions that encourage privacy harms and an overall degradation of obscurity.

Lawmakers and judges could prohibit combining facial recognition with existing surveillance systems designed for particular uses (like CCTV) or target "dark patterns," which are interface elements and designs that trick users into unwanted or unintentional exposures to facial recognition systems against their best interests.[55] The U.S. Federal

[50] See, *e.g.*, L. Taylor *et. al.* (eds.), *Group Privacy: New Challenges of Data Technologies*, 2017; D. Solove and W. Hartzog, *Kafka in the Age of AI and the Futility of Privacy as Control*, in *Boston University Law Review*, vol. 104, 2024, especially 1025-1029.

[51] S. Viljoen, *A Relational Theory of Data Governance*, in *Yale Law Journal*, vol. 131, 2021, 573.

[52] See, *e.g.*, J.A.T. Fairfield and C. Engel, *Privacy As A Public Good*, in *Duke Law Journal*, vol. 65, 2015, 387; D.K. Citron and D.J. Solove, *Privacy Harms*, in *Boston University Law Review*, vol. 102, 2022, 793, especially 831.

[53] See, *e.g.,* W. Hartzog and N. Richards, *The Surprising Virtues of Data Loyalty*, in *Emory Law Journal*, vol. 71, 2022, 985, especially 1012; W. Hartzog and N. Richards, *Legislating Data Loyalty*, in *Notre Dame Law Review Reflection*, vol. 97, 2022, 356; N. Richards and W. Hartzog, *A Duty of Loyalty for Privacy Law*, in *Washington University Law Review*, vol. 99, 2021, 961.

[54] W. Hartzog, *Privacy's Blueprint: The Battle to Control the Design of New Technologies*, Cambridge, MA, Harvard University Press, 2018, 279.

[55] See, *e.g.*, G. Gunawan *et al., Understanding Dark Patterns in Home IoT Devices*, in *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 2023 (forthcoming); R. Calo, *Digital Market Manipulation*, in *George Washington Law Review*, vol. 82, 2014, 995; G. Conti and E. Sobiesk, *Malicious Interface Design: Exploiting the User*, in *Proceedings of the WWW Conference*, 2010; L. Di Geronimo *et al*., *UI Dark Patterns and Where to Find Them: A Study on Mobile Applications and User Perception*, in

Trade Commission has filed complaints against companies for "unfair default settings," design choices that unfairly risk the security of personal data, and design choices that unfairly interfere with a technological privacy safeguard.[56]

Another way lawmakers might target the design of facial recognition systems is to expand theories of secondary liability to account for dangerous design choices. The FTC has developed a "means and instrumentalities" theory of wrongdoing for unfairly designing tools to encourage consumer harm. Product liability law has long developed theories of wrongdoing around design and warning defects. Given the widespread fraud and harassment enabled by mass facial recognition, these theories should be leveraged in combination with lawmakers' and judges' recognition of collective and social harms. All of these approaches—specific design rules, consumer protection doctrines, and expanded notions of secondary and product liability, can be leveraged to check the starting point for virtually any facial recognition system.

Finally, lawmakers should get serious about outright prohibitions and bans on facial recognition. Privacy law's favorite tool is procedure.[57] Surveillance laws justify observation through warrants and subpoenas. Data privacy laws justify information processing through consent or upon proof of certain contracts or business interests. People are given privacy when they have "control" over personal information and rights of transparency, access, and deletion.[58] Both due process and the Fair Information Practices, the bedrock principles of surveillance and data protection law, are built upon the idea that if you follow the right procedures, surveillance and data processing are justified.[59] The problem with procedural frameworks is that they end up justifying the practices they seek to mitigate.[60] This is a recipe for normalizing facial recognition because if the procedure is followed, the technology is used in a way that, over time, becomes unexceptional and entrenched.

Instead, lawmakers should outright substantively prohibit the dangerous activities that no amount of procedure can justify. Lawmakers might join cities like Portland, San Francisco, Oakland, Sommerville, and others that have banned facial and biometric surveillance by law enforcement or in places of public accommodation.[61] The European Union's AI Act, which prohibits facial surveillance and emotion recognition as part of a broad ban on unacceptably risky AI systems is the most prominent and substantive example of this approach.[62] Though even this

Proceedings of the CHI Conference on Human Factors in Computing Systems, 2020; C.M. Gray *et al.*, *End User Accounts of Dark Patterns as Felt Manipulation*, in *Proceedings of the ACM on Human-Computer Interaction*, vol. 5, n. 372, 2021, 1, especially 5; J. King and A. Stephan, *Regulating Privacy Dark Patterns in Practice—Drawing Inspiration from the California Privacy Rights Act*, in *Georgetown Law Technology Review*, vol. 5, 2021, 251; J. Luguri and L. Strahilevitz, *Shining a Light on Dark Patterns*, in *Journal of Legal Analysis*, vol. 13, 2021, 43.

[56] See, *e.g.*, *Complaint, Zoom Video Communications*, Inc., FTC Matter/File Number 192 3167, 9 November 2020; see also D. Solove and W. Hartzog, *The FTC Zoom Case: Does the FTC Need a New Approach?*, in *LinkedIn*, 2020, www.linkedin.com/pulse/ftc-zoom-case-does-need-new-approach-daniel-solove; R.Hutchinson, FTC says IntelliVision's claims about facial recognition software were deceptive | Opinion, Commercial Appeal (8 April 2025), www.commercialappeal.com/story/opinion/contributors/2025/04/08/better-business-bureau-ftc-facial-recognition/82983225007.

[57] See, *e.g.*, J.E. Cohen, *How (Not) to Write a Privacy Law,* in *Knight First Amendment Institute*, 2, 2021, 8, https://knightcolumbia.org/content/how-not-to-write-a-privacy-law/; A.E. Waldman, *Privacy, Practice, and Performance*, in *California Law Review*, vol. 110, 2022, 1221. The FTC also filed a complaint against Rite Aid for "Unfair Facial Recognition Practices," which involved using "facial recognition technology in their retail stores without taking reasonable steps to address the risks that their deployment of such technology was likely to result in harm to consumers as a result of false-positive facial recognition match alerts." Complaint, FTC v. Rite Aid Corp., Case No. 2:23-cv-5023, 19 December 2023.

[58] D. Solove and W. Hartzog, *Kafka in the Age of AI and the Futility of Privacy as Control*, in *Boston University Law Review*, vol. 104, 2024, especially 1025-1029; A.E. Waldman, *The New Privacy Law*, in *UC Davis Law Review Online*, vol. 55, 2021, 39-40.

[59] J. Rule *et al.*, *The Politics of Privacy*, Belmont, CA, Wadsworth Publishing, 1980, 93.

[60] See also J.E. Cohen, *How (Not) to Write a Privacy Law*, in *Knight First Amendment Institute*, 2, 2021, 8, https://knightcolumbia.org/content/how-not-to-write-a-privacy-law.

[61] See, *e.g.*, R. Metz, *Portland Passes Broadest Facial Recognition Ban in the U.S.*, in *CNN*, 2020, www.cnn.com/2020/09/09/tech/portland-facial-recognition-ban/index.html; S. Flynn, *13 Cities Where Police Are Banned from Using Facial Recognition Tech*, in *Innovation & Tech Today*, 2023, https://innotechtoday.com/13-cities-where-police-are-banned-from-using-facial-recognition-tech/ (last visited 2 March 2023).

[62] European Union AI Act, Chapter II, Article 5: "The following AI practices shall be prohibited:...e) …the use of AI systems that create or expand facial recognition databases through the untargeted scraping of facial images from the internet or CCTV footage; (f) …the use

supposed high-water mark for substantive bans on facial recognition has been criticized as not enough or even a form of privacy theater that functionally authorizes facial recognition in the most important contexts in spite of ostensibly prohibiting it.[63] The AI Act still permits the use of facial recognition for limited purposes by governments. While these purposes are laudable, such as finding missing persons, they are a foot in the door. Lawmakers must make fewer compromises if they are to protect against normalizing facial recognition and protect obscurity. Bright-line prohibitions on facial recognition provide a substantive backstop to prevent surveillance creep. In other words, they protect people by restricting dangerous behavior now

matter how acclimated people become to being watched through privacy nicks.

We believe society can't wait years for lawmakers and judges to get the balance just right when regulating facial recognition to maximize its benefits while minimizing its harms. By then, facial recognition infrastructure will be ubiquitous, and exploiting its full potential will seem like a good use of resources. And the accompanying surveillance creep such a balance would bring is inevitable. The law singles out specific technologies all the time because they are so exceptional. Automobiles, spyware, medical devices, and a host of other technologies have their own specific rules. Airplanes and telecommunications technologies were given their own federal regulatory agencies. Facial recognition is similarly worthy of exceptional treatment.

While outright prohibitions are more politically fraught and practically inflexible, they are the most significant tools available to resist the normalization of surveillance. It might sound extreme to call for an outright ban on the most dangerous surveillance practices, even when they might have some utility, but we think it is necessary.[64] Even if advocates of informed consent and warrant requirements for government searches got everything on their wish list, society would still end up worse off with facial recognition. We would suffer unacceptable harm to our obscurity and collective autonomy through a barrage of I agree buttons and search warrants powered by government and industry's unquenchable thirst for more access to our lives. There is only one way to sufficiently stop the harms of face surveillance. It must be banned outright.[65] Compromises that fall back on procedure and "individual control" will end up compromising the entire endeavor.

6. *Conclusion*

Facial recognition is truly a one-of-a-kind technology - and we should treat it as such. Our faces are central to our identities, online and off, and they are difficult to hide. People look to our faces for insight into our innermost feelings and dispositions. Our faces are also easier to capture than biometrics like fingerprints and DNA, which require physical contact or samples. And facial recognition technology is easy to use and accessible, ready to plug into police body cameras and other systems.

of AI systems to infer emotions of a natural person in the areas of workplace and education institutions, except where the use of the AI system is intended to be put in place or into the market for medical or safety reasons; (g) …the use of biometric categorisation systems that categorise individually natural persons based on their biometric data to deduce or infer their race, political opinions, trade union membership, religious or philosophical beliefs, sex life or sexual orientation; this prohibition does not cover any labelling or filtering of lawfully acquired biometric datasets, such as images, based on biometric data or categorizing of biometric data in the area of law enforcement; (h) the use of 'real-time' remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement, unless and in so far as such use is strictly necessary for one of the following objectives: (i) …the search for missing persons; (ii) the prevention of a specific, substantial and imminent threat to the life or physical safety of natural persons…; (iii) the localisation or identification of a person suspected of having committed a criminal offence….").

[63] Shubam, *EU AI Act Will Fail Commitment to Ban Biometric Mass Surveillance*, in *Reclaim Your Face* 2024, https://reclaimyourface.eu/eu-ai-act-will-fail-commitment-to-ban-biometric-mass-surveillance; Access Now, *The EU AI Act: A Failure for Human Rights, a Victory for Industry and Law Enforcement*, in *Access Now*, 2024, www.accessnow.org/press-release/ai-act-failure-for-human-rights-victory-for-industry-and-law-enforcement/.

[64] See, *e.g.*, E. Selinger and W. Hartzog, *The Inconsentability of Facial Surveillance*, 122 ("[I]f facial recognition becomes entrenched in the private sector by procedural frameworks, that means that in addition to a warrant framework's accretion problem, the government will also have a backdoor to retroactive surveillance via the personal data industrial complex. Through public/private cooperation, surveillance infrastructure will continue to be built, chill will still occur, harms will still happen, norms will still change, collective autonomy still will suffer, and people's individual and collective obscurity will bit by bit continue to diminish").

[65] *Id.*

The stakes are high. Over time, the slippery slope of normalizing facial recognition stands to change fundamental social beliefs about and dispositions toward privacy. The endpoint of the slope is the wide-scale degradation of obscurity protections necessary for pursuing the good life and maintaining the full potential of a liberal democracy. One of the most problematic aspects of society becoming acclimated to privacy nicks is that we become unable to fully appreciate how our autonomy, and thus dignity, are being routinely violated. We are being programmed not to worry about forms of face surveillance that once struck many of us as creepy, ambiguous threats. Over time, these privacy diminutions, once seen as worrisome, fail to trigger even basic concern. Lawmakers must not allow society to grow ever more alienated from appreciating the goods privacy offers without engaging in the oversight required to protect our invaluable obscurity.

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ORIGINAL PAPER

AI recruitment algorithms and the dehumanization problem

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Abstract
According to a recent survey by the HR Research Institute, as the presence of artificial intelligence (AI) becomes increasingly common in the workplace, HR professionals are worried that the use of recruitment algorithms will lead to a "dehumanization" of the hiring process. Our main goals in this paper are threefold: (i) to bring attention to this neglected issue, (ii) to clarify what exactly this concern about dehumanization might amount to, and (iii) to sketch an argument for why dehumanizing the hiring process is ethically suspect. After distinguishing the use of the term "dehumanization" in this context (i.e. removing the human presence) from its more common meaning in the interdisciplinary field of dehumanization studies (i.e. conceiving of other humans as subhuman), we argue that the use of hiring algorithms may negatively impact the employee-employer relationship. We argue that there are good independent reasons to accept a substantive employee-employer relationship, as well as an applicant-employer relationship, both of which are consistent with a stakeholder theory of corporate obligations. We further argue that dehumanizing the hiring process may negatively impact these relationships because of the difference between the values of human recruiters and the values embedded in recruitment algorithms. Drawing on Nguyen's (in: Lackey, Applied Epistemology, Oxford University Press, 2021) critique of how Twitter "gamifies communication", we argue that replacing human recruiters with algorithms imports *artificial values* into the hiring process. We close by briefly considering some ways to potentially mitigate the problems posed by recruitment algorithms, along with the possibility that some difficult trade-offs will need to be made.

Keywords Hiring Algorithms · Employee-Employer Relation · Stakeholder Theory · Gamification · Dehumanization · Human Resources

Introduction

As the presence of artificial intelligence (AI) becomes increasingly common in the workplace, Human Resources (HR) professionals are worried that the use of recruitment algorithms will lead to a "dehumanization" of the hiring process. In this paper, our aim is to examine this concern, which has received little attention in debates about the ethics of algorithms. We begin by discussing a recent survey of HR professionals, which reports current attitudes about the use of such AI-based tools in the workplace. In general, expectations are such that in the next few years, AI will play a prominent role in the HR toolkit, especially for hiring and onboarding purposes. Perhaps the most common objection to the use of hiring algorithms or algorithmic decision-making systems in general is that they have the potential to be biased or lead to objectionable patterns of discrimination. However, while HR professionals have registered concerns about bias and discrimination, interestingly the *most cited* worry in the recent survey that we consider is that hiring algorithms will "dehumanize" the hiring process. Our main goals in this paper are threefold: (i) to bring attention to this neglected issue, (ii) to clarify what exactly this concern about dehumanization might amount to, and (iii) to sketch an argument for why dehumanizing the hiring process is ethically suspect. After distinguishing the use of the term "dehumanization" in this context (i.e. removing the human

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presence) from its more common meaning in the interdisciplinary field of dehumanization studies (i.e. conceiving of other humans as subhuman), we argue that fears about dehumanizing the hiring process can be fruitfully investigated by considering its potentially negative impact on the employee-employer relationship. We argue that there are good independent reasons to accept a genuine, substantive employee-employer relationship, as well as an applicant-employer relationship, both of which are consistent with a stakeholder theory of corporate obligations. We go on to argue that dehumanizing the hiring process may negatively impact these relationships because of the difference between the values of human recruiters and the values embedded in recruitment algorithms. Drawing on Nguyen's (2021) critique of how Twitter "gamifies communication", we argue that replacing human recruiters with algorithms imports *artificial values* into the hiring process. We close by briefly considering some ways to potentially mitigate the problems posed by recruitment algorithms. However, as the use of recruitment algorithms becomes more widespread, difficult trade-offs may need to be made, since the human element, as it features in the hiring process, can also be the source of its own problems.

HR managers react to the idea of hiring algorithms

In 2019 the HR Research Institute[1] published a report entitled "The 2019 State of Artificial Intelligence in Talent Acquisition."[2] This report was based on a survey of 484 responses from HR professionals across a wide variety of industries and includes representatives from both small businesses and large firms. According to the report "many HR professionals foresee AI playing enhanced roles in most aspects of talent acquisition" over the next few years (3). There are many ways in which AI may contribute to HR's traditional functions, e.g. chatbots to answer employee questions and to guide recently hired employees through the onboarding process. But according to *recruitment* professionals, what they desire most from the use of AI is to "find more qualified candidates" and "to reduce the time they spend sifting through resumes" (4). One reason for interest in utilizing AI for recruitment is that only about 25% of survey respondents rated their talent acquisition methods very favorably (6). Even though currently only 10% of survey respondents claimed that their company uses AI to a high or very high degree in the hiring process, according to the report, 36% of respondents claimed that in the next two years they expect their company to begin making heavy use of AI for recruitment, and 61% expect to see at least moderate usage (11).

While survey respondents expect that *every* stage of the hiring process will be impacted by AI in the next few years, some stages are expected to be more affected than others. The first stage of the recruitment process consists of *sourcing* a pool of potential candidates. For lucrative positions, a firm might not need to do much advertising to attract applicants. Still, we can imagine that AI-based tools might be used to conduct *targeted searches* to come up with a preliminary pool of potential candidates (14–18). The second stage of the hiring process involves *screening* potential candidate to narrow down what might be a rather large pool. Here, AI-based tools might be used to quickly process and prioritize resumes or to predict the fit between a candidate's personality and the organization's "culture" (19–29). The third stage of the hiring process is the *interview stage*, where candidates are evaluated more closely by the hiring committee. Here, AI-based tools might be used to automate the first round of interviews or to contribute to the evaluation of candidates (30–4). The last stage of the hiring process is the *selection stage*, where an offer is extended to the successful applicant. Here, AI-based tools might influence the final selection process by recommending the best candidates in the remaining pool (34–7). Of course, even if computational algorithms directly impact the selection stage, it will prove some time before AI comes close to having the final say in hiring decisions. In general, survey respondents expect AI to play the largest role in the earlier phases of the recruitment process, e.g. sorting through resumes to establish a preliminary pool, since these are the facets of the hiring process that depend less on sophisticated human judgment calls, and so are more amenable to automation (12).

To get a better sense of how AI is currently impacting the hiring process, consider the empirical study by Raghavan et al. (2019) of 18 vendors offering AI-based recruitment services that are currently being employed by HR professionals at the *screening* stage. Raghavan et al. identify various assessment types that are used by these vendors, including personality tests, situational judgment tests, gameplay (e.g. puzzle-solving), and video interview analysis (2019, p. 6). These assessment tools can be either customized to meet the needs of the client, or they can be pre-built and general-purpose. In conjunction with "training data" provided either by the vendor or the employer, proprietary algorithms are used to make predictions about the expected degree of success of a prospective employee. For example, HR might provide one of these AI recruitment vendors with data on past employees and request a

[1] The HR Institute is the research division of HR.com, the largest online community for HR professionals.

[2] https://www.oracle.com/a/ocom/docs/artificial-intelligence-in-talent-acquisition.pdf?elqTrackId=1279a8827f3d4548ae3f966beeeef458&elqaid=83,148&elqat=2.

report that predicts the expected sales numbers or performance reviews of the prospective employee based on certain features about them. To do so, the vendor might use computational algorithms to analyze a short video of the candidate that detects desirable personality traits thought to be correlated with success, such as "openness", "enthusiasm", and "warmness". As Raghavan et al. note, these "video-based assessments, in particular, are increasingly common" (2019, p. 12). Indeed, the practice has become widespread enough to draw the attention of lawmakers. In 2019 Illinois, USA passed the *Artificial Intelligence Video Act*, which requires employers to inform and obtain consent from applicants if their video interviews will be analyzed by some AI-based assessment tool.

The dehumanization problem

The most cited ethical concern in the HR research institute report

Many scholars across a wide array of disciplines have raised a host of ethical concerns about the use of predictive analytics algorithms for making decisions that greatly affect the life prospects of human beings (Mittelstadt et al., 2016). Perhaps the most common objection that the use of such technology prompts is that it already is or has the potential to be *biased* or *unfair,* leading to objectionable patterns of discrimination (e.g. Eubanks, 2018; Noble, 2018). A particularly high-profile example of a biased hiring algorithm was one built by Amazon (but allegedly never used), which systematically discriminated against resumes from women applicants (Dastin, 2018). While HR managers have registered worries about bias and discrimination, interestingly, *the most cited* ethical concern about the use of hiring algorithms in the HR Research Institute report is the "dehumanization of the recruitment process" (49). Unfortunately, the report does not elaborate much on how we should understand "dehumanization" in this context. Remarks along similar lines include the concern that the use of AI for hiring purposes will make the recruitment process lose its "human touch" (49). This ethical concern about the possibility of dehumanizing the hiring process is interesting because it does not depend on the outcome of debates over whether hiring algorithms are biased. Future technical and ethical advances may address problems of bias and discrimination (Kleinberg et al., 2020), but worries about dehumanization seem to be largely independent of these other issues. Since the worry about dehumanizing the hiring process was the ethical concern that was most widely cited in this survey of HR professionals, it is worthwhile to attempt to get clearer on the nature of the problem.

Distinguishing two distinct meanings of "dehumanization"

As Smith (2018, p. 264) aptly observes, "mentioning dehumanization raises the question of what dehumanization is. That question is surprisingly difficult to answer." In trying to make sense of the worry that hiring algorithms will dehumanize the recruitment process, one might attempt to investigate the phenomenon in light of the interdisciplinary field of dehumanization studies, which spans several disciplines including philosophy, political science, social psychology, disability studies, history, etc. The modern origin of dehumanization studies is often traced back to scholarly reactions to the atrocities of the twentieth century, especially Hannah Arendt's work on the social mechanisms by which the Holocaust was perpetrated (Kronfeldner, 2021, p. 3). Given that the field of dehumanization studies is "rather patchy" (Kronfeldner, 2021, p. 1), it comes as little surprise that there are many different accounts of dehumanization in the literature. As Smith (2016, pp. 418–19) helpfully points out, there are at least eight different ways in which "dehumanization" has been conceptualized by various scholars:

(i) Subjecting others to indignities; or, in a more Kantian vein, treating them merely as means.
(ii) Verbally likening others to nonhuman animals or inanimate objects.
(iii) Denying the subjectivity, individuality, agency, or distinctively human attributes of others.
(iv) Denying that others undergo mental states.
(v) Treating others in such a way as to erode, obstruct, or extinguish some of their distinctively human attributes.
(vi) Conceiving of others as inanimate objects.
(vii) Conceiving of others as less human than members of one's ingroup.
(viii) Conceiving of others as subhuman creatures.

The sense of "dehumanization" that Smith favors and elaborates upon (e.g. 2014, 2016, 2018) is that which is suggested by (viii), conceiving of others as subhuman, or somewhat paradoxically, as both human and subhuman simultaneously. Many of the other senses of dehumanization listed above are related to and often go hand-in-hand with the process of conceiving of others as subhuman, but the latter, as Smith argues, is what is essential to dehumanization (2016, p. 419).

Although there are many different ways in which dehumanization might manifest itself, the idea that dehumanization, at bottom, involves the conception of other humans as less than human and thus lacking moral standing, captures the core element of dehumanization. As Maria Kronfeldner observes in her valuable introduction to the *Routledge*

Handbook of Dehumanization, despite its many complexities, in cases of dehumanization "one aspect seems to be always present: dehumanization establishes *difference* and *distance* between human beings (2021, p. 9). For example, the atrocities inflicted on the Jewish people and other targeted groups by the Nazis during World War II "are often taken as paradigmatic cases of dehumanization", and of course the Nazis explicitly conceived of the victims of their brutality "as less human if not less than human" (Kronfeldner, 2021, p. 14). The ability to dehumanize other human beings, in the sense of conceiving of them as subhuman in this way, is made possible, as Smith argues, by "our propensity for essentialistic and hierarchical thinking" (2018, p. 268).

Unfortunately, as our brief sketch of standard accounts of dehumanization makes clear, the scholarly literature on dehumanization will not be of much help in clarifying the central concern that HR managers have about hiring algorithms. Despite the use of term "dehumanization" in the HR Research Institute report, it is implausible that survey respondents have in mind the standard sense of dehumanization whose essence involves conceiving of other human beings as subhuman. Clearly, nothing so extreme as the phenomenon that dehumanization scholars investigate is liable to occur in the case of hiring algorithms. Moreover, given that this form of dehumanization is often associated with grave human rights violations and even genocide, it would be offensive to even make that comparison. But if we cannot understand the qualms that HR managers have about hiring algorithms by reference to the most prominent accounts of dehumanization, then one might be suspicious of the concerns raised by the HR Research Institute report. More specifically, one might worry that the term "dehumanizing" is being employed here, as is often the case in conversations about new technologies, as "an unspecific negative qualifier—used in order to stress a negative evaluation of whatever phenomenon is under study" (Kronfeldner, 2021, p. 17). This worry is particularly pressing given that questions about whether new technologies are dehumanizing have been denounced by some critics as "largely meaningless" because, it is claimed, terms like "dehumanizing", when not referring to the phenomenon investigated in dehumanization studies, are "not well defined" (Mowshowitz, 2008, p. 281).

While we think that the worries that HR managers have cannot be subsumed under the most prominent accounts of the phenomenon investigated in the dehumanization studies literature, we do not think these concerns should be dismissed so easily. Certainly, the claim that hiring algorithms have the potential to be dehumanizing is not literally *meaningless*. The sense of dehumanization at play in the case of hiring algorithms does not involve conceiving of other humans as subhuman, but instead involves removing the human presence from some process or domain, where previously the human presence was taken as a given. This sense of dehumanization is not unique to the burgeoning debate over hiring algorithms. We can find similar uses of the term "dehumanization" in the debate over "lethal autonomous weapons systems", e.g. drone warfare. For instance, after acknowledging that the term "dehumanization" can refer to the "war victims' sufferings", Joerden (2018, p. 56) notes "there is a meaning of the term 'dehumanization' that must be set apart, namely, that mankind withdraws, so-to-speak, from the immediate war-related events, and thus, 'dehumanization' of war occurs." We might also refer to this alternate meaning of dehumanization succinctly as "leaving out the human" (Kronfeldner, 2021, p. 17). Henceforth, this is the account of dehumanization we will have in mind. With this alternative sense of the term "dehumanization" in place, it is clear that hiring algorithms would leave out the human, and thus it is clear that hiring algorithms would dehumanize the hiring process.

The central question now though is whether there is anything wrong with leaving out the human in this particular context. It requires little argument to show that conceiving of and treating other humans as subhuman is always seriously morally wrong. Indeed, this more common sense of dehumanization investigated by Smith and others seems to be a negatively valenced instance of what Bernard Williams (1985) would call a "thick ethical concept." These are concepts that already have evaluative content "built into" them. The common sense of "dehumanization" clearly already has wrongness built into it. However, as we have argued, this is not the sense of dehumanization at issue in the case of hiring algorithms, which instead involves removing the human presence. Moreover, this alternative sense of dehumanization is *not* a thick ethical concept; it does not have wrongness built into it. In some cases, it is morally permissible to leave out the human, e.g. having a robot vacuum the floor rather than a human. The problem then though is that once we distinguish these two different senses of the term "dehumanization," it is not at all obvious that dehumanizing the hiring process, in the sense of removing the human presence, is morally objectionable. In the following sections, we aim to give a novel argument, drawing on some related discussions in the ethics of technology, for why dehumanizing the hiring process in this alternative sense *should* be regarded as ethically suspect.[3]

[3] We think our formulation of this alternative sense dehumanization in terms of leaving out the human is sufficiently clear for our purposes, but we do not pretend that this is a completely unproblematic conceptual analysis. For example, some questions about how much the human presence must recede in order to count as dehumanizing the process will remain. Even in the debate on lethal autonomous weapons systems, humans have withdrawn a considerable degree from the process, but of course, not entirely so. Presumably, how much withdrawal is sufficient to count as a dehumanized process will depend on the context.

A threat to the employee-employer relationship

The shareholder theory versus the stakeholder theory

A fruitful approach for better understanding why dehumanizing the hiring process might be ethically suspect is to consider how the use of recruitment algorithms might negatively impact the *employee-employer relationship*. Now, those in the grips of Friedman's (1970) infamous "shareholder theory", which states that the sole obligation that corporate executives have is to run the firm in accordance with the wishes of the shareholders, which "generally will be to make as much money as possible while conforming to the basic rules of the society", might balk at the idea of a genuine, substantive employee-employer relationship. The shareholder theory, however, has been the object of much criticism, which continues into the twenty-first century (e.g. Aglietta & Rebérioux, 2005; Stout, 2012). As a result, an alternative to Friedman's model of corporate governance has gained prominence in subsequent decades—a view often referred to as the "stakeholder theory" (e.g. Freeman, 1984).[4] Contra Friedman, the stakeholder theory states that the corporation has obligations to *all stakeholders*, whether or not this maximizes profit. In the widest sense, stakeholders include any groups or individuals "who benefit from or are harmed by, and whose rights are violated or respected by, corporate actions" (Freeman, 1984, p. 100). Typically, this includes owners, employees, suppliers, contractors, customers, investors, local communities, the media, the government, etc.

Nowadays, it is becoming increasingly common for corporate executives to at least pay lip service to the stakeholder theory. In August 2019, the "Business Roundtable" released a statement signed by almost 200 hundred corporate executives, including the CEOs of Amazon, Apple, JPMorgan Chase, Disney, etc., explicitly disavowing previous shareholder-centric principles of corporate governance, and instead pledging "commitment to all of [their] stakeholders.".[5] With respect to employees specifically, the signers pledge to: (i) compensate employees fairly and provide them with important benefits, (ii) support them with training and education to develop new skills, and (iii) promote diversity, inclusion, dignity, and respect in the workplace. Since the stakeholder theory does not subordinate all corporate actions and values to profit maximization, the theory allows for a richer and more robust conception of the ideal employer-employee relationship.

Individuating and distinguishing distinct social relationships

Beyond its connection to the stakeholder theory, there are good independent reasons to accept a substantive conception of the employee-employer relationship. In an influential discussion of the value of privacy, James Rachels argues that privacy is important for "our ability to create and maintain different sorts of social relationships with different people" (1975, p. 326). In doing so, Rachels offers us some helpful clues for how to establish the reality of distinct social relationships. As Rachels observes, distinct social relationships are associated with different patterns of behavior; for example, "a man may be playful and affectionate with his children…but businesslike with his employees, and respectful and polite with his mother-in-law" (ibid.). Moreover, the information that it is appropriate for a person to share depends on the nature of the social relationship at stake. Crucially, these distinct patterns of behavior, along with what information it is appropriate for others to have, are also that which *individuate* and *distinguish* distinct social relationships. As Rachels (1975, p. 327) points out, "different patterns of behavior are (partly) what define the different relationships; they are an important part of what makes the different relationships what they are." Although there may be some cultural and individual variation in one's conception of these distinct relationships, "there is inseparable from that conception an idea of how it is appropriate to behave with and around them, and what information about oneself it is appropriate for them to have" (Rachels, 1975, p. 329). Indeed, more specific discussions of privacy and the employee-employer relationship presuppose something akin to Rachels' general thesis that norms about the sharing of information are inextricably connected to the distinct nature of the social relationship at issue (Simms, 1994; Superson, 1983).

The upshot for our purposes is that we know that there is a genuine, substantive employee-employer relationship because we know that there exist here different patterns of behavior and different standards of privacy, when compared to, say, a friendship or the generic benefactor-to-beneficiary relationship. In light of the dominance of the stakeholder theory of corporate obligations and the fact that there are good independent reasons to believe in a genuine, substantive employee-employer relationship with its own distinct norms, we have good reason to care about the potential impact of hiring algorithms on the relationship between employees and their employers.

In addition to the employee-employer relationship, the foregoing considerations should also make us concerned

[4] For a more recent discussion of the stakeholder theory, see Freeman et al. (2018).

[5] See: https://opportunity.businessroundtable.org/ourcommitment/.

about the relationship between employers and applicants. As is the case with other distinct social relationships, unique patterns of behavior and what information it is appropriate to share are associated with the applicant-employer relationship. Since, typically, at most a few candidates are hired from a pool of sometimes hundreds of applicants, it is noteworthy that the majority of those who will be affected, and potentially wronged, by hiring algorithms in any *given* search will be applicants. Thus, if we take a stakeholder perspective, we should care about the impact of hiring algorithms not just on the employee-employer relationship, but also on the relationship between applicants and employers. For convenience, we will speak of the "employee-employer" relationship, but our argument will apply, *mutatis mutandis*, to the prospective-employee-employer relationship.

Artificial values in recruitment algorithms

In the section "The dehumanization problem", we drew attention to the primary concern that HR managers have with recruitment algorithms: the worry that such a hiring method would *dehumanize* the recruitment process. In the section "A threat to the employee–employer relationship", we proposed to understand ethical qualms about dehumanizing the recruitment process (i.e. removing the human presence) as fears about negatively impacting the employee-employer relationship, although it is still unclear exactly what sort of violation recruitment algorithms would constitute. In this section, we aim to further elaborate on why it is that dehumanizing the recruitment process by the use of hiring algorithms poses a threat to the employee-employer relationship.

Gamification and artificial value creation

To understand how the use of hiring algorithms has the potential to negatively impact the employee-employer relationship, we can draw inspiration from criticisms of other, similar moral conundrums related to new and emerging technologies. In his paper "How Twitter Gamifies Communication" C. Thi Nguyen (2021) argues that Twitter "gamifies" communication between people, changing the nature of their discourse. By "gamifies communication", Nguyen (2021, p. 411) specifies that Twitter offers, "immediate, vivid, and quantified evaluations of one's conversational success" in a way that allows the user to improve upon past "performances" (tweets) to score more "points" (likes, retweets, and followers). Of course, gamification is not always bad—in fact, it is often a good thing, allowing us to complete tasks more easily and more enjoyably. As Nguyen points out, (2021, pp. 413–414), we can use game-based learning to teach ourselves healthy habits such as exercise and proper nutrition. For example, devices like *FitBit* allow users to track their health progress in ways that reward them for meeting fitness goals; apps such as *Calm* even gamify relaxation, by rewarding the user for participating in exercises that are meant to induce physical and mental relaxation.

The problem with the gamifying *sincere* interpersonal communication though ultimately stems from how games motivate participation. Crucially, Nguyen argues, games draw us in by *simplifying* and thus *clarifying* our desires: "Gamification increases our motivation by changing the nature of the activity. Often, the goals of ordinary activity are rich and subtle. When we gamify these activities, we change those goals to make them artificially clear" (2021, p. 411). For example, when engaged in non-gamified *sincere* discourse with our peers, we have complex and multifaceted values for this complex and multifaceted good, as (presumably) ought to be the case. But to "succeed" at Twitter, one must engage in discourse with an *artificial* value set—the set given to them by the site creators—and focus on the goal of the game: maximizing engagement with their tweets. There may be many tasks which could be unproblematically gamified, in virtue of the fact that we don't believe we ought to hold rich, subtle, complex values for them. For instance, toothbrushes which gamify oral hygiene for children may fall into this category.

For Nguyen, values are "artificial" when they are a part of a structure of agency that has been given to us by someone else. In games, someone is the game designer who "sculpts the temporary agency that the player will occupy during the game", determining "who the player will be" in the imagined world of the game (Nguyen, 2021, p. 414). Artificial values, then, are those values that are part and parcel of the entire agential structure that we are given—our end goals, our character, the ways in which we are to move and act, etc.—either by the designer of the game (if we are playing a game) or by some other mechanism that suspends our usual agency.

Furthermore, such suspensions of autonomous agency grant the same artificial values to all persons whose agency is suspended by the same person or mechanism. This is what allows for multi-player games and what allows Twitter to function as efficiently as it does. But for people to share values, Nguyen argues, these values must be simpler and clearer than our ordinary values are outside the context of a pre-sculpted agency. In real life, our values are often opaque to us, and figuring out how to balance them against one another (e.g. "I want to relax tonight but I also want to support my friend by going to her art show") is anything but clear. But in a game, our values are perfectly clear, which allows us to effectively strategize. Nguyen uses the example of our values for "likes', "retweets", and "followers" handed to us by the site designers when we enter the "game" of

Twitter. Of course, outside of Twitter, where our agency is not "sculpted" for us, we still value being liked and admired by our peers. But valuing peer admiration, popularity, or even fame, is not the same thing as valuing Twitter likes or retweets, which are specific discrete digital events. On Twitter, interaction with one's tweets is the entire point of tweeting, and tweets are designed and hashtagged with maximum engagement as the goal. Outside of "game play", our value for peer recognition is simply one value among a host of innumerable social values, and maximum attention is *not* the goal we generally have when interacting with others in the real world outside of Twitter.

Similarly, in board games like tic-tac-toe, we adopt an artificial value for getting three X's or O's in a row. These are artificial values because i) they are values we do not hold outside of the game, and ii) they are simple and clear—we are never conflicted about whether we should try to attain three X's or O's in a row. In general, when we engage in a game or a gamified activity that is well-structured, we are never in the dark about how important certain goals or achievements are. Our artificial values are crystal clear to us because the game designer hands them to us, perfectly packaged. This is what makes game-play fun instead of laborious. "Gamifying" an otherwise difficult or boring task, therefore, is an attempt to "impose value clarity on a pre-existing thicket of values" (Nguyen, 2021, p. 415).

Of course, there is a clear difference between the gamification of Twitter and the ways in which recruitment algorithms are currently being employed by some firms. For one thing, Twitter intentionally gamifies communication by handing the user a set of artificial values that will keep them hooked on the site. Recruitment algorithms usually do not intentionally gamify the hiring process, although there are notable exceptions, as in the case of game-based assessments discussed above. Our primary argument in this section is *not* that recruitment algorithms may dehumanize the hiring process by gamifying it. We introduce the concept of gamification only because this is how Nguyen introduces his concept of artificial values. But artificial values may enter into an activity even if that activity has not been fully gamified, and this, we will argue, makes the use of recruitment algorithms *prima facie* morally objectionable.

Artificial values and recruitment algorithms

Although gamification of the hiring process per se does not entirely justify concerns about dehumanization, we believe both gamification and the use of AI recruitment algorithms share a common worrisome feature. Our proposal here is that the same problem that Nguyen sees with gamified communication arises in the case of offloading hiring responsibilities to recruitment algorithms, namely, the creation of *artificial values* in human-to-human relationships. Recall Nguyen's critique of Twitter's gamification of communication is that it involves a sort of trade: our normal, complex, opaque, human values, in exchange for easy artificial ones. When we play games, or are involved in gamified activities, our practical agency is ruled by these artificial values, rather than our normal values that we act on in our real lives. The concern in the case of gamified communication is that the complexity of how and why we value human communication *ought not* to be simplified.

In the employment context, how these artificial values arise is rather simple: algorithms are quantifiers, and as such will make hiring decisions or recommendations through a process of quantification based on the values fed into the system by the programmer. As we discussed in the section "HR managers react to the idea of hiring algorithms", the vendors that sell these assessment tools are quick to assure us that this quantified predictive process is incredibly complex and extremely accurate, based on careful validation studies, etc. Still, as AI currently exists, even the most complex algorithmic values are not identical to human values—they are artificial, created by the human programmer. Something that the use of AI-based tools for recruitment purposes highlights is that human relationships look nothing like that which holds between a person and a computer algorithm. This is, in part, because the values of the recruitment algorithm are never affected by the actual human behind the resume. In the context of human relationships, we expect to be appreciated in the sort of way humans appreciate other humans, rather than be understood through a quantified assessment of individual bits of personal data.

To better understand the concern with AI algorithm values replacing the real human values in hiring situations, it will help us to imagine a very value-sensitive recruitment scenario. Imagine a ballet dancer who has been training since childhood to audition for the New York City Ballet. As she sits in the overflow room, waiting for her turn to audition, she is informed that the judges will only see two more dancers before they have to go home for the evening, and that she will not be one of those two. She is told, however, not to worry. The judges have purchased a highly advanced algorithmic decision-making system that collects millions of points of data on all remaining contenders, scans film of their dances in search of skills that predict future success, and even checks their family history for relatives who have struggled with substance abuse or obesity. The dancer is heartbroken. But why?

There are two reasons the dancer may be dismayed to learn that her dancing will be judged by an algorithm rather than the human judges she had anticipated. The first reason is that she may feel she has less chance of success being judged by the algorithm than did the dancers who performed for the human judges. We anticipate many readers will connect with this feeling, though it is hard to put a finger on

exactly why this is. Why should our ballet dancer *not* be relieved to be performing for such an *impartial* judge, capable of assessing her raw talent without being distracted by biases or illicit expectations? The answer seems to be that we are often more confident in our ability to communicate important aspects of ourselves to humans rather than to computers. Most of us have been inculcated with skills of human interaction which run deeper than any purely computational or theoretical analysis could explain; we are able to read people, inspire people, reassure and calm people, and communicate with people through untold numbers of subconscious movements, facial expressions, verbal and non-verbal vocal cues, along with too many others to list here. With AI we are, to put it simply, out of our element. It is not how we naturally communicate, and this puts us at a disadvantage.

The second reason is more along the lines of our primary critique of recruitment algorithms: the dancer may worry that, even if she gets hired, the victory will be hollow. Rather than being chosen because her dancing truly and deeply moved the judges, she will have gotten the job because thousands of details about her were quantified and compared to the traits of others, and hers came out on top. While she may be comparably happy to be hired, she is unhappy about how and why she is hired, because she feels reduced to her attributes, disjointed, and objectified. Or, perhaps more accurately, she feels cut up into small bits of data, unsure whether herself as whole dancer—a whole person—is truly wanted as an employee of the ballet. Already we can see here a fraught element introduced into the would-be employer-employee relationship.

There is a more precise way of describing what seems morally objectionable about auditioning for an algorithm. Like Nguyen's critique of Twitter's gamification of communication, we suggest that in the case of the ballet dancer, the values placed on the skills of the dancers are given an *artificial clarity* by an algorithm that is programmed to do the job of the human judges. No matter how hard the programmer tries, the values of the algorithm can never really be like the values of those particular judges. No matter how hard *she* tries, the ballet dancer's dancing will not—cannot—be valued by the AI in the same way that it would be valued by human judges. It is not, to be clear, merely in virtue of the program's status as AI that it cannot value the dancer in the same way that a human judge would. The AI's values are artificial because i) they are given to the AI by another entity (its programmer) and ii) because they enjoy a clarity that distinguishes their metric of weighting and comparing from the opaque metric of regular human values. Nguyen elaborates on this concern as follows:

> In ordinary life, our values are hard to balance. I care about spending time with my loved ones, raising my children right, writing good philosophy, enjoying myself in rock climbing, staying healthy, and eating delicious food. Not only are my values often in tension, but there is usually no way to precisely compare them. (2021, p. 415)

This seems right—any attempt to articulate precisely how our values affect each other, weigh and pull against one another to guide our everyday lives, will be, strictly speaking, a falsehood. If the judges value things like grace, strength, facial expressiveness, fluidity, genuineness, and earnestness, it is anyone's guess how exactly each of these values will be weighted in the individual performers. For any two dancers with equal amounts of grace and earnestness, the earnestness of one may put her ahead of the other for no other reason than that the judges value her *personal earnestness* in ways too complex and opaque to ever elaborate on. Likewise, AI recruitment algorithms rely entirely on artificial value trade-offs that rob our human relationships of what makes them distinctively valuable—perhaps, that our human relationships are not very much like a precisely-programmed algorithm at all. There are obviously disanalogies between the job of a ballet dancer and the sort of job AI recruitment algorithms would normally be recruiting for—but as we see it, the difference is one of degree, not kind.[6]

Is the employee–employer relationship worth saving?

The sorry state of the employee–employer relationship

Thus far we have been motivated to better understand the most widely cited concern among HR managers regarding the increasing use of predictive algorithms in the hiring process. More than anything else, HR managers are worried that the process of identifying candidates, screening and interviewing them, and onboarding them, will lose the "human touch" if conducted by AI. As we have proposed, one reason to view this sort of dehumanization (i.e. removing the human presence) as ethically suspect is that the artificial values imported by hiring algorithms may be a threat to the employee-employer relationship, in the sense that the relationship is stripped of something characteristic of the ways in which humans typically engage with one another:

[6] Our argument in this section should be viewed as part of an ongoing conversation about some of the detriments of advanced technology. This conversation was spearheaded in the twentieth century by some classical critiques of technology, industrialized society, and "instrumental rationality", often associated with authors belonging to the Frankfurt School, such as Marcuse (1941), Adorno and Horkheimer (1972/2020), and Habermas (1970). See Schecter (2010) for a general overview.

the complex, opaque ways in which the values appear and interact in the relationship. But is the employee-employer relationship worth saving? One may object that, even if hiring algorithms posed a threat to the employee-employer relationship in this way, the relationship is already fraught with so many non-ideal aspects that improving—or preventing further damage to—the relationship would be wasted effort.

To be sure, business ethicists and political philosophers alike have long bemoaned the sorry state of the contemporary employee-employer relationship, even before the rise of AI in the workplace. A point that Borowski observed over two decades ago would ring true to many employees today, namely that the relationship between the employee and the employer "seems to be an adversarial one—almost as if management and workers were on two different competing teams" (1998, p. 1624). Indeed, this is precisely how Joseph Jones, a worker in an Alabama Amazon warehouse recently described his experience: "It's a very adversarial relationship with the supervisors and the staff…You're a cog in the system…and it's very obvious" (Sherman, 2021). An adversarial relationship of this sort is liable to lead to feelings of tension, distrust, and antagonism (Borowski, 1998, pp. 1625–6).[7] In a similar fashion, Karnes (2009, p. 189) laments the "disintegration of employer-employee relationships." According to Karnes (2009), gone are the days when workers and employers trusted each other (190), when both parties demonstrated commitment to each other's well-being (191), when employees and employers genuinely felt a sense of unity and togetherness (192), etc. These impressions are supported by some empirical data. According to the 2014 "Work and Well-Being Survey" conducted by the *American Psychological Association*, "employee distrust is pervasive in the U.S. workforce."[8] More specifically, only 50% of workers believed that their employer is "open and upfront" with them, and about 25% of workers claimed that they do not trust their boss.

Some modest proposals

To counteract the growing sense of distrust, tension, and antagonism between employees and employers, Borowski proposes injecting a dose of Kantian ethics into the workplace: both workers and employers need to treat each other with the respect that any human person deserves in virtue of being a rational and autonomous creature (1998, p. 1626–7). A similar solution put forward by Karnes consists, largely, in putting the stakeholder theory into practice: employers should take into account workers' basic human needs, seek to create enthusiastic employees, treat their workers fairly and justly, recognize the genuine accomplishments of their employees, and foster genial and cooperative relations in the workplace (2009, pp. 194–5). In general, employers and employees ought to treat each other with the decency and respect that each party is owed in virtue of being human, regardless of profit margins (Karnes 2009, p. 195). We might take some inspiration from these proposals in addressing the issue of using hiring algorithms in the workplace. Insofar as employers opt to use algorithms for hiring purposes, then they should, as far as possible, be guided by these broadly Kantian ideals of respect for persons *as such*, which may involve limiting the use of hiring algorithms to, say, the *screening stage*. This is not to say that employers must themselves be Kantians. Such a principle is available to those holding a wide array of normative views, including ethical pluralism (Ross, 1930), utilitarianism (Mill 1864/2014), virtue ethics and care ethics (Kittay, 2005). Most of these frameworks take respect for persons as a *prima facie* value, though different frameworks will have different reasons for doing so.

In our view then, even though the state of the employee-employer relationship may already be quite bad, this does not mean that we ought not to care about the impact of the use of hiring algorithms on the relationship between employees and their employers. To the extent that HR managers are themselves concerned about dehumanization, we have a *prima facie* reason to understand and address the worry. Furthermore, insofar as we take the stakeholder theory of corporate responsibility seriously, corporations may very well have a social responsibility to reduce the extent to which employees and prospective employees experience the negative effects of hiring algorithms in their relationship with the firm. If we are right that dehumanizing the hiring algorithms will negatively impact the employee-employer relationship, then these effects are cause for concern, even if they are not decisive.

Difficult trade-offs

However, it is worth considering, and indeed even plausible, that avoiding the ills of dehumanizing the hiring process will always come at the expense of (some amount of) objectivity or impartiality. Imagine that our ballet dancer from before manages to audition before human judges rather than an algorithm. Furthermore, suppose her performance entirely captivates her human audience, some of whom are moved to tears over the nearly preternatural grace of her rendition of Odette's dance of death from Swan Lake's fourth act.

[7] Relatedly, Elizabeth Anderson (2017) famously defends the view that most American workers are subject to an illiberal "private government" in the workplace.

[8] See: https://www.apa.org/news/press/releases/2014/04/employee-distrust. Similar results were obtained by the American Psychological Association's more recent 2017 survey on employee well-being. See: https://www.apaexcellence.org/assets/general/2017-work-and-wellbeing-survey-results.pdf.

Our dancer is hired. The judges perhaps cannot put their fingers on exactly why her performance was so magnificent, but it likely has something to do with how each detail came together as a unique whole, unable to be reduced to specificities. The judges were moved, and they know this dance will move their audiences as well. Yet, the capacity of humans to be *moved* by other humans is the same capacity that allows bias and prejudice to enter in. Suppose our dancer had performed an identical dance for a different set of judges—judges who had each had a recent bad experience with someone who looked vaguely like our dancer. This historical fact works an emotional effect on the judges; they are, unbeknownst to themselves, less open to being moved by her dance, and so her performance leaves them feeling cold. The dancer is not hired.

Importantly, such concerns may be understood as legitimate worries about *humanization* in the hiring process. It is the *human* element, not the machine element per se, which internalizes bias and prejudice toward women, ethnic minorities, queer or gender non-conforming individuals, religious minorities, or other marginalized groups. So, it is at least worth considering whether efforts at avoiding bias in the hiring process must necessarily come at the expense of the "human element" of the relationship. We should be open to the possibility that there is a clash of values ideals. Our goal in this paper is to elucidate concerns about dehumanizing the hiring process, which—once we disambiguate this sense of dehumanization (i.e. removing the human presence) from the more common sense of dehumanization (i.e. conceiving of others as subhuman)—is not obviously morally wrong. However, it is quite possible that minimizing bias may, at times, require forfeiting some of the "humanity" of the hiring process and thus negatively impacting the employee-employer relationship. Likewise, it is probably the case that working to re-humanize the process would increase the risk of bias and prejudice. Although we lack the space to fully address this concern here, it is plausible that such trade-offs need to be decided on a case-by-case (company-by-company, job-by-job, etc.) basis. Perhaps, in the end, there is no general principle for perfectly balancing these conflicting ideals (e.g. Ross, 1930).

Conclusion

The use of AI-based tools for hiring new employees is increasing in popularity, despite some unease among HR managers. While many discussions in the ethics of algorithms are concerned with problems of bias and discrimination, in this paper we have sought to examine the issue of recruitment algorithms with an eye toward the under-explored concerns of these HR managers—in particular, the concern that the use of such technologies will "dehumanize" the hiring process. As we have argued, this sense of dehumanization, which simply involves "leaving out the human", should be carefully distinguished from the more common sense of the term, as it features in dehumanization studies, according to which dehumanization involves conceiving of others as subhuman. Clearly, nothing of the sort happens in the case of recruitment algorithms. Still, as we have argued, there is something ethically suspect about removing the human presence from the hiring process. Our argument for being suspicious of the use of hiring algorithms is that they are unable to truly take the place of a human recruitment officer, as these algorithms rely on *artificial values* that only approximate but never replicate real human values. In human relationships, our values and the relations between our values are often complex and difficult to balance. Any attempt to impose artificial value clarity on this cascade of values by means of predictive algorithms—even ones that have been subjected to rigorous validation methods—is bound to strike some employees and employers as uncomfortably alienating, thereby negatively impacting the employee-employer relationship. This concern is something that HR managers must reckon with as the presence of AI in the workplace increases in the coming years.

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